LAB INSTRUCTIONS

Lab 3 – Numerical Methods for Unconstrained Optimization

Objectives

- Implement first- and second-order numerical optimization algorithms from scratch.
- Apply the conjugate gradient method to unconstrained optimization problems.
- Apply the modified Newton's method and compare convergence behavior.
- Analyze convergence in terms of optimal point, function value, and iteration count.
- Gain practical understanding of algorithmic optimization beyond symbolic analysis.

Part A – Conjugate Gradient Method

Write a program in MATLAB or Python that implements the **conjugate gradient method** for unconstrained optimization.

Using your implementation, solve the following exercises from Arora:

- Problem 10.52
- Problem 10.54
- Problem 10.56

Your code should follow the algorithmic structure presented in the course textbook (see recommended reading below).

Part B – Modified Newton's Method

Write a program in MATLAB or Python that implements the **modified Newton's algorithm** for unconstrained optimization.

Using your implementation, solve the same set of problems:

- Problem 10.52
- Problem 10.54
- Problem 10.56

Compare the behavior of the modified Newton method to the conjugate gradient method in terms of convergence speed and robustness.

10.52 $f(x_1, x_2) = x_1^2 + 2x_2^2 - 4x_1 - 2x_1x_2$; starting design (1, 1)

10.54 $f(x_1, x_2) = 6.983x_1^2 + 12.415x_2^2 - x_1$; starting design (2, 1)

10.56 $f(x_1, x_2) = 25x_1^2 + 20x_2^2 - 2x_1 - x_2$; starting design (3, 1)